

Agostino Scalzullo

Audio Software & Creative Technology Engineer C++ · DSP · JUCE · Max/MSP · Audio ML · Deep Learning · Real-Time Systems · Computer Vision

London, UK · OSTIN SOLO · Founder, VSTOPIA contact@ostinsolo.co.uk · [Live Portfolio](#) · [GitHub](#)

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Profile

Audio software and creative-technology engineer working across native C++, DSP, plug-ins, neural audio, computer vision, networked audio and interactive hardware. Founder of VSTOPIA and developer of commercial OSTIN SOLO music-production software, with experience taking experimental systems through native implementation, testing, deployment, licensing and distribution.

Core Skills

Native audio: C/C++, Objective-C++, JUCE, Max SDK, VST3/AU, DSP, MIDI/MPE **ML / deep learning / vision:** source separation, BS-RoFormer, Demucs, MossFormer2, ECAPA speaker embeddings, target-speaker extraction, MLX, Core ML, CLAP embeddings, Depth Anything, MediaPipe, ARKit **Realtime / platform:** WebRTC, Opus, RTP MIDI, WebSockets, AVFoundation, hardware APIs, Swift/SwiftUI, macOS/iOS **Engineering:** profiling, benchmark design, parity/regression testing, model/runtime packaging, Developer ID signing and notarization **Cross-platform builds:** macOS Apple Silicon & Intel, Windows x64/ARM64, cross-architecture toolchains, conditional compilation and release packaging

Selected Work

DSm — Dynamic Split Module

Neural separation, restoration and web-sampling environment for Ableton Live, created by combining and extending the earlier Split Wizard+ and YouTube 4 Live projects, with 63 model workflows spanning Demucs, RoFormer and related architectures. Covered by Sound On Sound and listed in the official Cycling '74 project ecosystem. [Product](#) · [Sound On Sound](#) · [Cycling '74](#)

Native RoFormer External & VST3/AU — Work in Progress

Native C++/MLX implementation bringing six-stem BS-RoFormer separation into a Max/MSP external and an in-development plug-in architecture. Across tested Apple Silicon workloads, optimisation achieved a **median 1.85x speedup**, reaching **2.5x on MelBand RoFormer**; native six-stem BS-RoFormer achieved **RTF $\approx$ 0.36 (~2.8x real time)** on M4 Pro with bit-exact parity against the Python MLX reference.

AUDIOENCE

JUCE/C++ networked collaboration plug-in architecture combining real-time audio, MIDI, WebRTC/P2P communication, Opus/PCM modes, resampling and DAW synchronisation across macOS and Windows. [Official project](#) · [Technical manual](#)

Native Max / MPE / Hardware Systems

Native C++ Max externals for multitouch, MPE, computer sensors, Depth Anything and AprilTag workflows. Includes Objective-C++ bridge layers, platform APIs, native model/runtime integration and ongoing JUCE VST3/AU migration. [Public product catalogue](#)

Real-Time Target-Speaker Voice Isolation — Private R&D

Real-time target-speaker extraction using MossFormer2 + ECAPA, with speaker enrollment and suppression of competing voices.

SoundCloner — Private R&D

Native synthesis and audio-analysis research platform combining deterministic DSP, parameter/genome representations, similarity scoring, candidate retrieval, ML inference and large synthetic evaluation infrastructure.

Sample-Space Exploration — Private R&D

CLAP/audio embeddings, classifier training, similarity retrieval and Three.js spatial exploration of audio corpora, supported by benchmark and evaluation tooling.

iOS Computer Vision — Private R&D

Swift/SwiftUI iOS application work using ARKit, SceneKit, AVFoundation and MediaPipe Tasks Vision for real-time facial/eye landmark processing and guided physical-makeup workflows with simulator and physical-device validation.

Engineering Validation

Repeatable benchmarking and regression methodology across projects: controlled baselines, latency/throughput profiling, held-out ML evaluation, parity testing, hardware-in-the-loop checks and cross-machine validation across Apple Silicon, Intel Mac and Windows x64/ARM64 execution paths where supported.

Experience

Founder & Audio Software Developer — VSTOPIA | 2023–Present

Built an independent software ecosystem spanning product development, web/backend infrastructure, distribution, licensing, updates and developer workflows.

Independent Audio Software Developer — OSTIN SOLO | 2018–Present

Commercial and experimental audio technology across DSP, Max for Live, native Max externals, plug-ins, machine learning, computer vision, MPE/hardware control and networked music systems.

Founder, Artistic Director & Experience Designer — Macrowave / Cacao Discoteca / Basement | 2014–2016

Directed electronic-music and multimedia programmes in Italy for audiences up to 3,000, spanning artist curation, production, projection mapping, scenography and interactive installations.

Education

Music Production — SAE Institute | 2019–2020 · Recording, editing, mixing, mastering, sound design and synthesis **Spanish Language Studies — Valencia | 2022 Electronic Music & Musicianship — Italian Conservatory | c. 2004–2009 Scientific High School Diploma — Liceo Scientifico, Italy | c. 2007–2012**

Languages

Italian — Native · English — Full professional proficiency, IELTS certified · Spanish — Professional working proficiency

Recognition & Public Evidence

Sound On Sound coverage of DS_m · Cycling '74 official project listing · OSTIN SOLO commercial product catalogue · public AUDIOENCE platform and technical documentation.

Full technical portfolio: <https://ostinsolo.github.io/ostin-solo-portfolio/>